IH Used Paper BTECH-11-11-20

R13

Code No: 115EH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, October - 2020
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) What are the different types of operating systems? Explain them in detail.
b) What are the main characteristics of Real Time Operating System? [8+7]

2.a) Describe in detail about computer system architecture.
b) What is a system call? Explain in detail the different types of system call. [7+8]

3.a) List out the differences between preemptive and non-preemptive scheduling.
b) State and explain critical section problem with suitable examples. [7+8]

4. Following is the snapshot of a CPU:

Process	CPU Burst	Arrival Time
P1	75	0
P2	40	10
P3	25	10
P4	20	80
P5	45	85

Draw the Gantt chart and calculate the turnaround time and waiting time of the jobs for FCFS (First Come First Served), SJF (Shortest Job First), SRTF (Shortest Remaining Time First) and RR (Round Robin with time quantum 15) scheduling algorithms. [15]

5.a) What is a Virtual Memory? Discuss the benefits of virtual memory technique.
b) Explain the concept of demand paging in detail. [8+7]

6. What are the similarities and differences between a segment and a page? Which one is transparent and which one is translucent to the user and why? [15]

7.a) Briefly explain about single-level, two-level and Tree-Structured directories.
b) Describe the disk performance parameters with examples. [8+7]

8. Write short notes on:
a) Capability based systems
b) Language based protection. [7+8]

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